

Non-Technical Summary

The completed code automates the asphalt mixture analysis process for NDOT's Balanced Mix Design workflow. Instead of requiring engineers to manually copy data into spreadsheets and calculate results by hand, the program reads the provided Excel data, organizes it, performs the required calculations, and creates clear output tables and graphs. The goal of the code is to make the testing process faster, more consistent, and less likely to have human error.

The program is divided into four main sections. The first section analyzes aggregate gradation data by combining the different aggregate materials based on the selected blend percentages. It then compares the combined gradation to NDOT's specification limits and creates a graph showing whether the mixture falls within the acceptable range.

The second section calculates volumetric properties of the asphalt mixture. It uses the lab measurements from the spreadsheet to calculate the important mixture properties and checks whether the air voids are within the required range. If the air voids are outside the acceptable limits, the code flags the sample as invalid for performance testing.

The third section analyzes IDEAL-CT cracking test data. It uses the load and displacement data from the test to create a curve, find important values from the curve, and calculate the CTIndex. This helps evaluate how well the asphalt mixture resists cracking.

The fourth section analyzes Hamburg Wheel Tracking Test data. This part of the code checks how the asphalt mixture performs under repeated wheel loading by tracking rut depth over time. It identifies the point where rutting behavior changes, determines whether the sample reaches the failure threshold, and decides whether the mixture passes or fails based on the required number of cycles.

Overall, the code turns a manual Excel-based process into an automated Python workflow. It reads the input data, completes the engineering calculations, checks the results against NDOT requirements, and produces useful tables and plots. This makes the Balanced Mix Design evaluation easier to repeat, easier to review, and more reliable for future asphalt mixture testing.